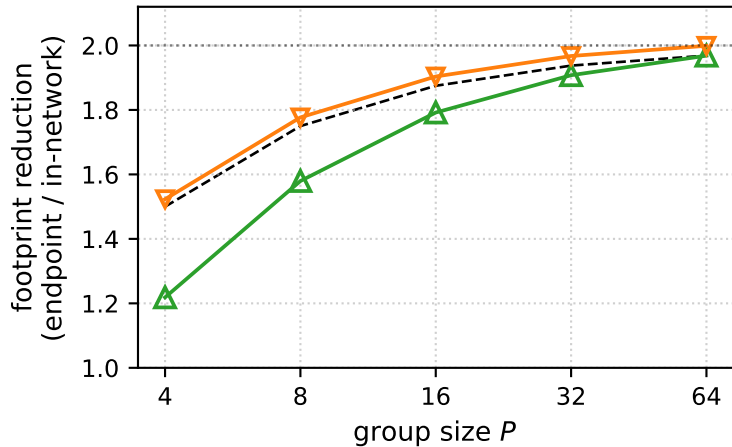
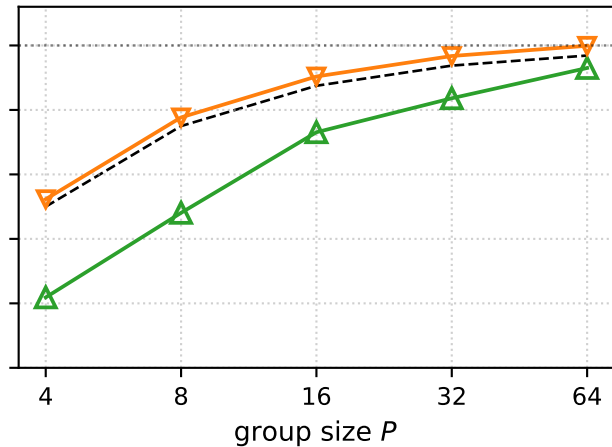


(a) single switch, $d = 1$



(b) three-tier, $d = 3$



Reduce-Scatter



AllGather



$2 - 2/P$